Exploratory Data Analysis (EDA) & Business Intelligence Report

1. Introduction

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on retail sales data, identify patterns, trends, and business insights, and develop key performance indicators (KPIs) and a business intelligence dashboard for decision-making.

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2. Dataset Overview

Dataset Information

- Dataset Name: Sample Superstore Dataset
- Domain: Retail Sales

Dataset Statistics

- Total Rows: 9,994
- Total Columns: 21
- Missing Values: 0
- Duplicate Records: 7

Main Columns

- Order ID
- Order Date
- Customer Name
- Product Name
- Category
- Region
- Sales
- Profit
- Quantity

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3. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Handled missing values.
- Converted date columns into datetime format.
- Validated numerical columns.
- Created clean_sales.csv for further analysis.

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4. Descriptive Statistics

Sales Statistics

- Mean Sales: 229.86
- Minimum Sales: 0.44
- Maximum Sales: 22,638.48
- Standard Deviation: 623.25

Profit Statistics

- Mean Profit: 28.66
- Minimum Profit: -6,599.98
- Maximum Profit: 8,399.98
- Standard Deviation: 234.26

Key Observations

- Sales values show significant variation across products.
- Profit distribution contains both positive and negative values.
- Some products generate high revenue but lower profit margins.

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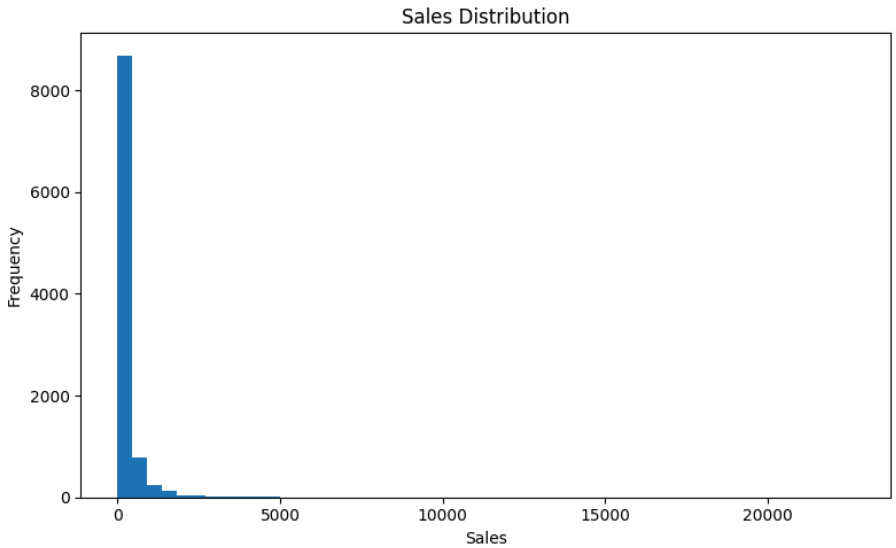
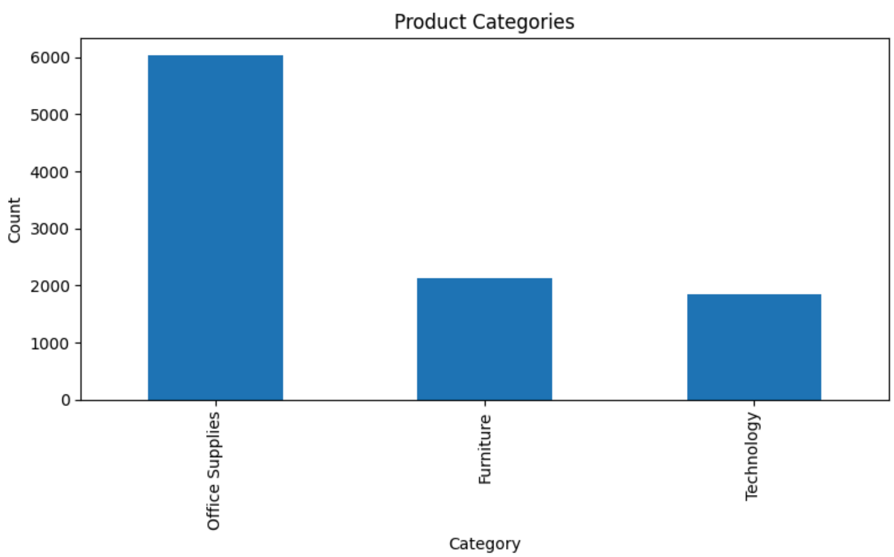
5. Visualizations

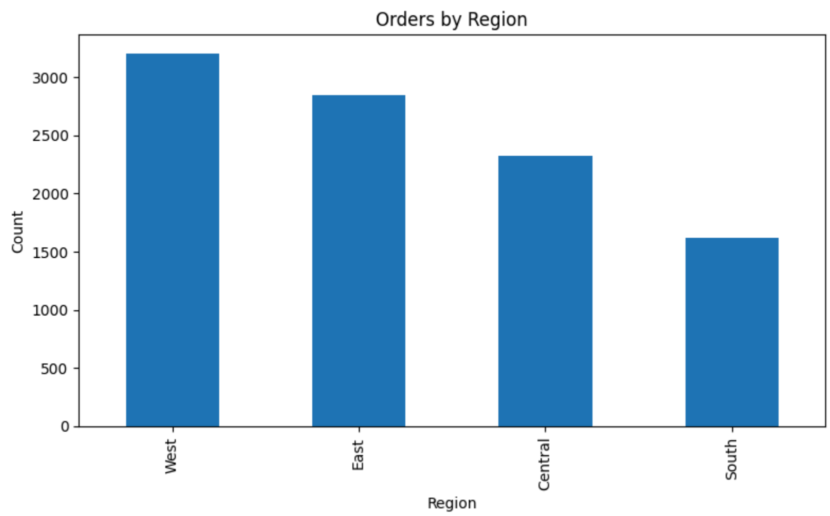
Sales Distribution Histogram

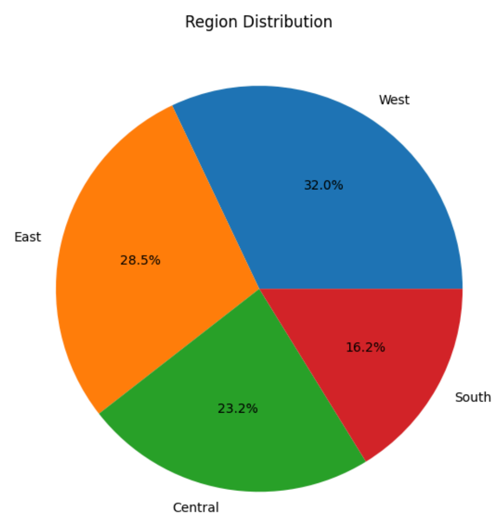


Category Analysis



Region Performance





Findings

- Technology and Office Supplies contribute significantly to sales.
- Regional performance varies across the dataset.
- Sales distribution is right-skewed.

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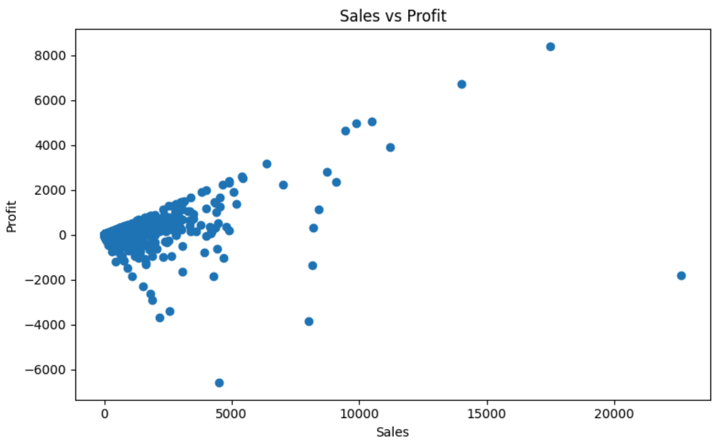
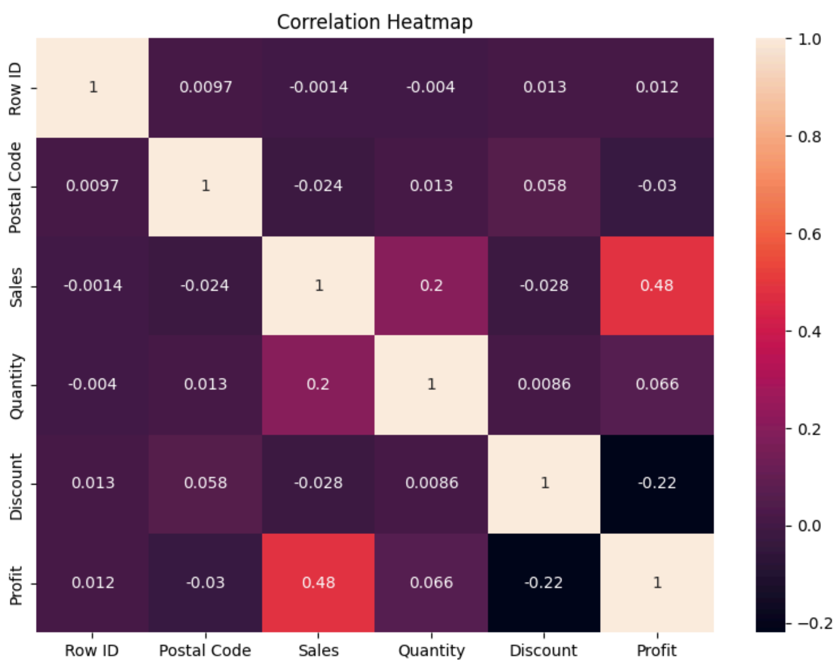
6. Multivariate Analysis

Scatter Plot: Sales vs Profit



Correlation Heatmap



Correlation Matrix

	Row ID	Postal Coc	Sales	Quantity	Discount	Profit
Row ID	1	0.009671	-0.00136	-0.00402	0.01348	0.012497
Postal Coc	0.009671	1	-0.02385	0.012761	0.058443	-0.02996
Sales	-0.00136	-0.02385	1	0.200795	-0.02819	0.479064
Quantity	-0.00402	0.012761	0.200795	1	0.008623	0.066253
Discount	0.01348	0.058443	-0.02819	0.008623	1	-0.21949
Profit	0.012497	-0.02996	0.479064	0.066253	-0.21949	1

Findings

- Sales and Profit show a positive relationship.
- Quantity has a weaker relationship with Profit.
- Strong correlations help identify important business drivers.

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7. KPI Summary

Sales KPIs

- Total Revenue: 2,297,201
- Total Orders: 5,009
- Total Customers: 793
- Average Order Value: 458.62

Profit KPIs

- Total Profit: 286,397
- Profit Margin %: 12.47%

Product KPIs

- Best Product: Canon imageCLASS 2200 Advanced Copier
- Top Category: Technology

Regional KPIs

- Best Region: West

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8. Dashboard

Power BI Dashboard

[Insert Dashboard Screenshot]

Dashboard Components:

- Revenue KPI Card
- Profit KPI Card

- Orders KPI Card
- Customer KPI Card
- Monthly Sales Trend
- Top Products Analysis
- Category Revenue Analysis
- Region Performance Analysis
- Profit Analysis
- Interactive Filters

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9. Conclusion

The EDA and Business Intelligence analysis revealed valuable business insights.

Key findings include:

- Technology category generated the highest revenue.
- The strongest-performing region contributed the largest share of sales.
- Sales and Profit demonstrated a positive correlation.
- KPI analysis highlighted the most profitable products and regions.
- The Power BI dashboard provides an effective way to monitor business performance and support data-driven decisions.

This project demonstrates an end-to-end analytics workflow involving data cleaning, exploratory analysis, SQL querying, KPI development, and dashboard creation.